GGG-S-116C
May 11, 1992
SUPERSEDING
GGG-S-116B
April 5, 1961

FEDERAL SPECIFICATION

SCRAPERS, SHIP, (PAINT AND SCALE REMOVING)

This specification is approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. Scrapers covered by this specification are hand tools for removing paint and scale from finished surfaces.

1.1.1 Federal specification coverage. Federal specifications do not cover all varieties of the commodity indicated by the title of the specification, or which are commercially available, but are intended to cover only those generally used by the Federal Government.

1.2 Classification.

1.2.1 Types and classes. Ship scrapers shall be of the following types and classes, as specified (see 6.1).

Type I - All metal
  Class A - Angle bent end
  Class B - Curved bent end
Type II - Deck, wood handle
Type III - Hull, wood handle
Type IV - Ship bottom, wood handle
Type V - Ships, triangular, wood handle

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

PPP-P-40	- Preservation and Packing of Hand Tools; Tools and Tool Accessories for Power Driven, Metal and Woodworking Machinery

Federal Standards:

FED-STD-376	- Preferred Metric Units for General Use by the Federal Government

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and Commercial Item Descriptions as outlined under General Information in the Index of Federal Specifications, Standards and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

(Single copies of this specification, and other Federal specifications and Commercial Item Descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston, MA; New York, NY; Philadelphia, PA; Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Houston, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.)

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(Federal Government activities may obtain copies of Federal standardization documents and the Index of Federal Specifications, Standards and Commercial Item Descriptions from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of military specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Non-Government documents. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM):

E 18 - Metallic Materials Rockwell Hardness and Rockwell Superficial Hardness of

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

3. REQUIREMENTS

3.1 Illustrations. The illustrations shown herein are descriptive and not restrictive provided the scrapers are otherwise in accordance with this specification.

3.2 Material.

3.2.1 Steel. Steel shall be as specified hereinafter for each type. When specified, the supplier shall furnish certification of the carbon content (see 4.2.3 and 6.1).

3.2.2 Wood handles. Wood handles shall be of good quality straight-grained hickory, hard maple, or tough white ash.

3.3 Hardness. At a midpoint within 1/2 inch back from scraping edges, the blade shall show a hardness value of not less than 45 nor more than 52 on the Rockwell C scale, when tested as specified in 4.3.1.

3.4 Marking. The ship scrapers shall be marked with the manufacturer's name or identifying symbol and the state or country of manufacture. The marking shall be engraved, molded, etched, or stamped in such a manner that it will remain clear and legible throughout the life of the item.

3.5 Finish.

3.5.1 Steel finish. Steel shall have either a painted, lacquered, or natural finish, at the option of the contractor, except for the scraping edges which shall be ground. Unless otherwise specified, all surfaces having a natural or ground finish shall be completely coated on delivery with a preservative suitable for the prevention of rust or corrosion in storage without adversely affecting the material coated.

3.5.2 Wood handle finish. Unless otherwise specified in the invitation for bids, wood handles shall be either with or without a protective coating of varnish, lacquer, shellac, or wax, at the option of the contractor.

3.6 Type I, class A, all metal scrapers, angle bent end. Type I, class A scrapers shall be similar to figure 1.

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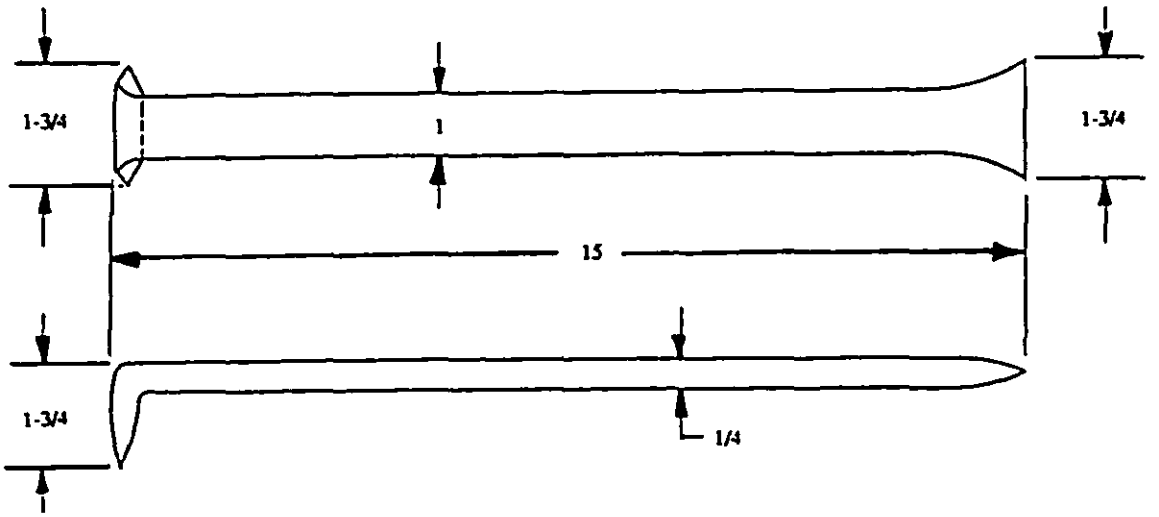


Figure 1. Type I, class A, all metal scraper, angle bent end.

3.6.1 Type I, class A manufacture and design. The scrapers shall be made from one piece of steel having a carbon content of 0.80 to 1.05 percent. Both ends of the tool shall have scraping edges, one of which shall be in alignment with, and the other at approximately right angles to, the longitudinal axis of the scraper body. Both ends shall be forged, hardened, tempered, and ground to a double-beveled scraping edge, and shall be satisfactory for the purpose intended.

3.6.2 Type I, class A dimensions. The scrapers shall comply with the dimensions given on figure 1 except that the dimensions may vary within a tolerance of ± 5 percent.

3.7 Type I, class B, all metal scrapers, curved bent end. Type I, class B scrapers shall be similar to figure 2.

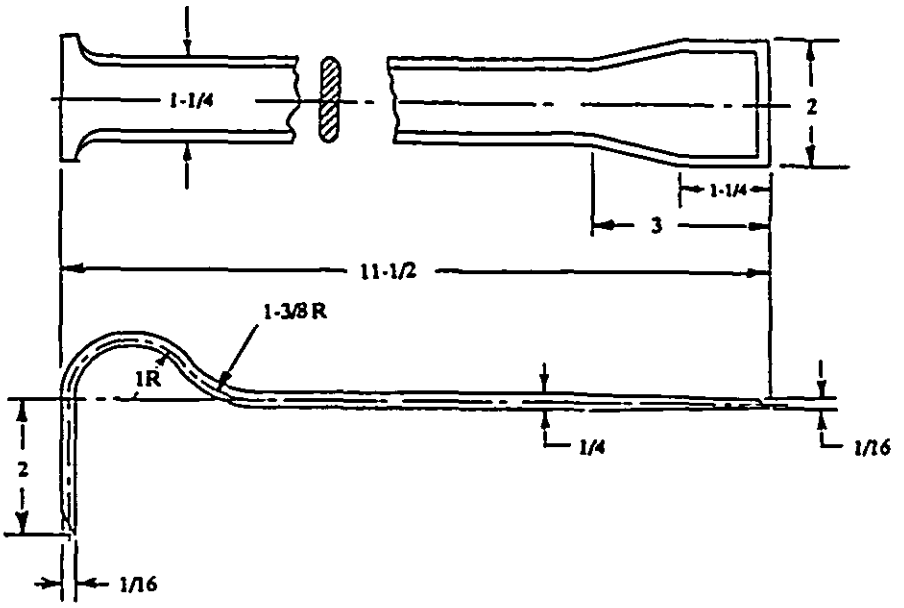


Figure 2. Type I, class B, all metal scraper, curved bent end.

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3.7.1 Type I, class B manufacture and design. The scrapers shall comply with the requirements of 3.6.1 except that the two scraping edges shall be ground to a single-beveled edge.

3.7.2 Type I, class B dimensions. The scrapers shall comply with the dimensions given on figure 2 except that the dimensions may vary within a tolerance of ± 5 percent.

3.8 Type II, deck scrapers, wood handle. Type II scrapers shall be similar to figure 3.

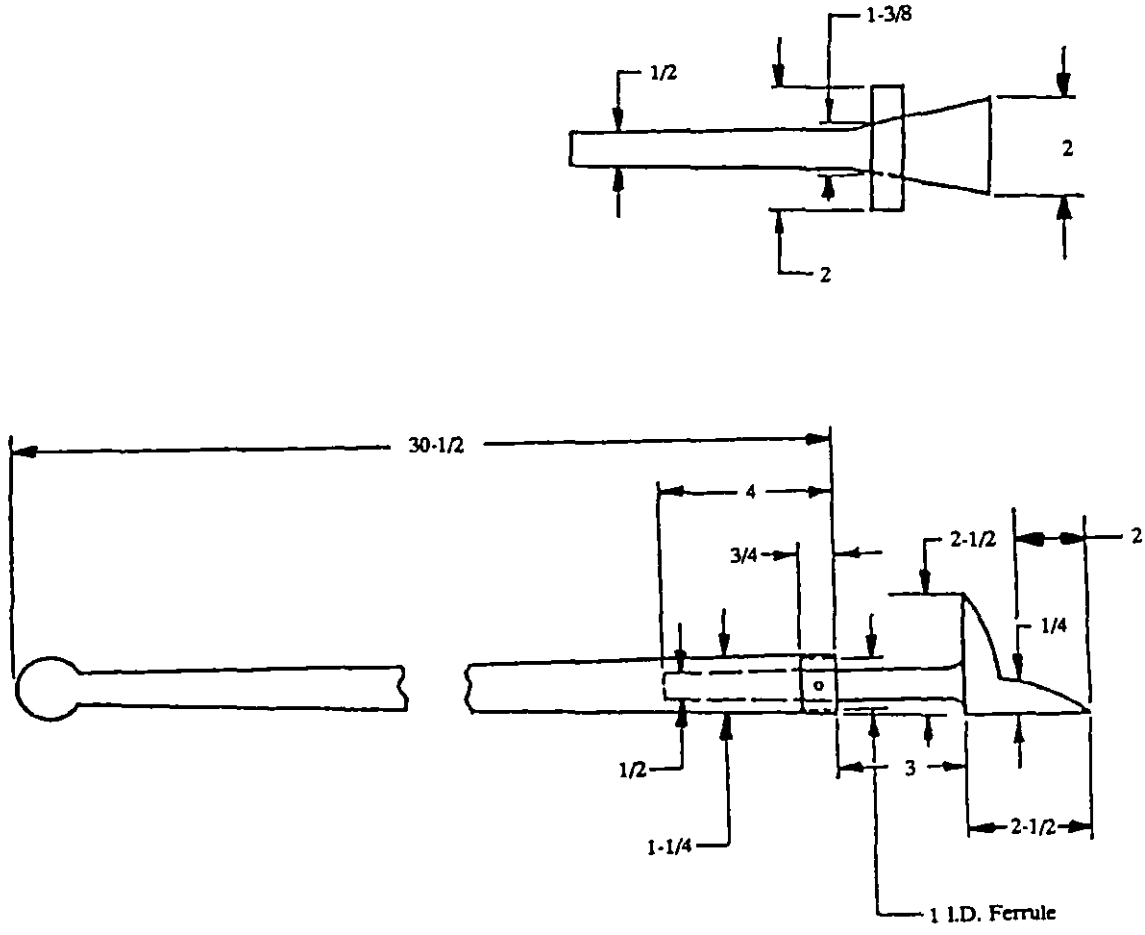


Figure 3. Type II, deck scraper, wood handle.

3.8.1 Type II, manufacture and design. The scrapers shall have a blade made from steel having a carbon content of 0.80 to 0.90 percent. The blade shall have two scraping edges located 90 degrees apart and properly hardened, tempered, and ground; one of the edges shall be in longitudinal alignment with the handle and the other edge at approximately right angles to the handle. The blade shall be either forged integrally with, or securely fastened to, a steel shank which shall fit securely into a wood handle. A substantial metal ferrule and a through rivet shall be provided to hold the shank securely in the handle under service conditions.

3.8.2 Type II handle. The handle shall be well proportioned and of suitable shape and size to provide a comfortable handgrip.

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3.8.3 Type II dimensions. The scrapers shall comply with the dimensions given in figure 3 except that the dimensions may vary within a tolerance of ± 5 percent.

3.9 Type III, hull scrapers, wood handle. Type III scrapers shall be similar to figure 4.

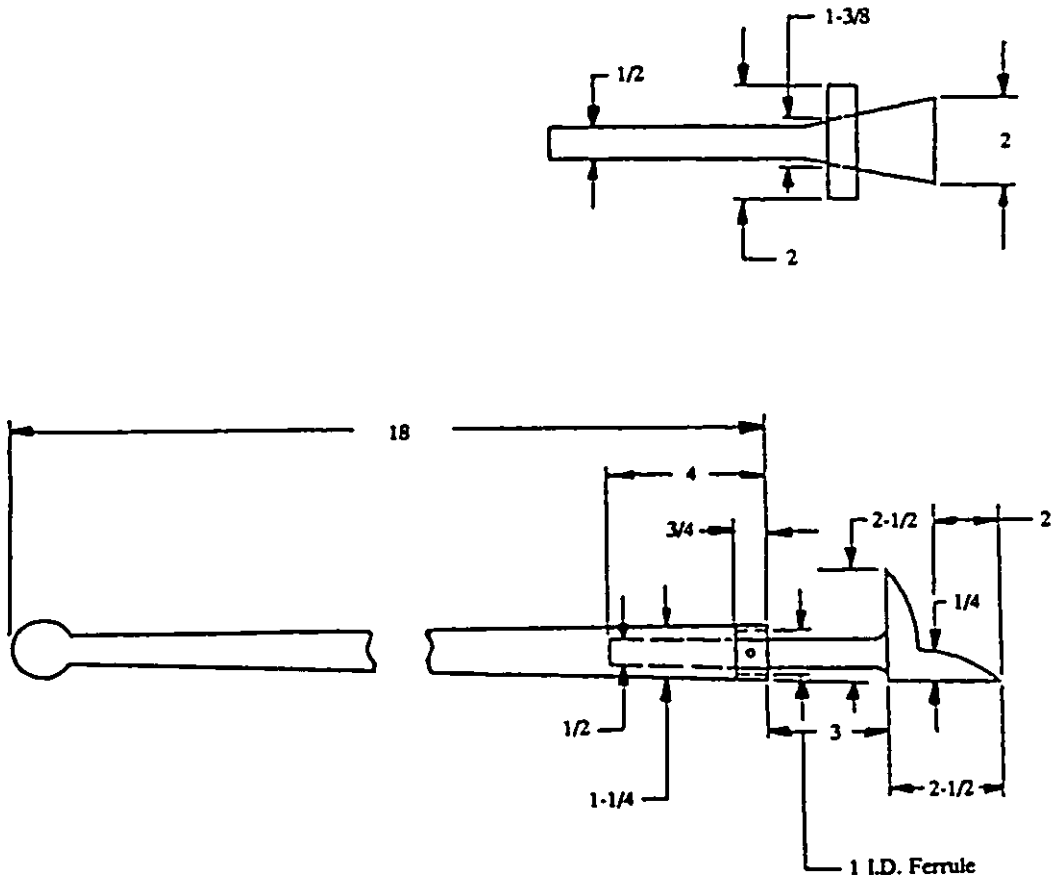


Figure 4. Type III, hull scraper, wood handle.

3.9.1 Type III manufacture and design. Type III scrapers shall be as specified for type II deck scrapers except that the handle length shall be 18 inches, subject to a tolerance of ± 5 percent.

3.10 Type IV, ship bottom scrapers, wood handle. Type IV scrapers shall be similar to figure 5.

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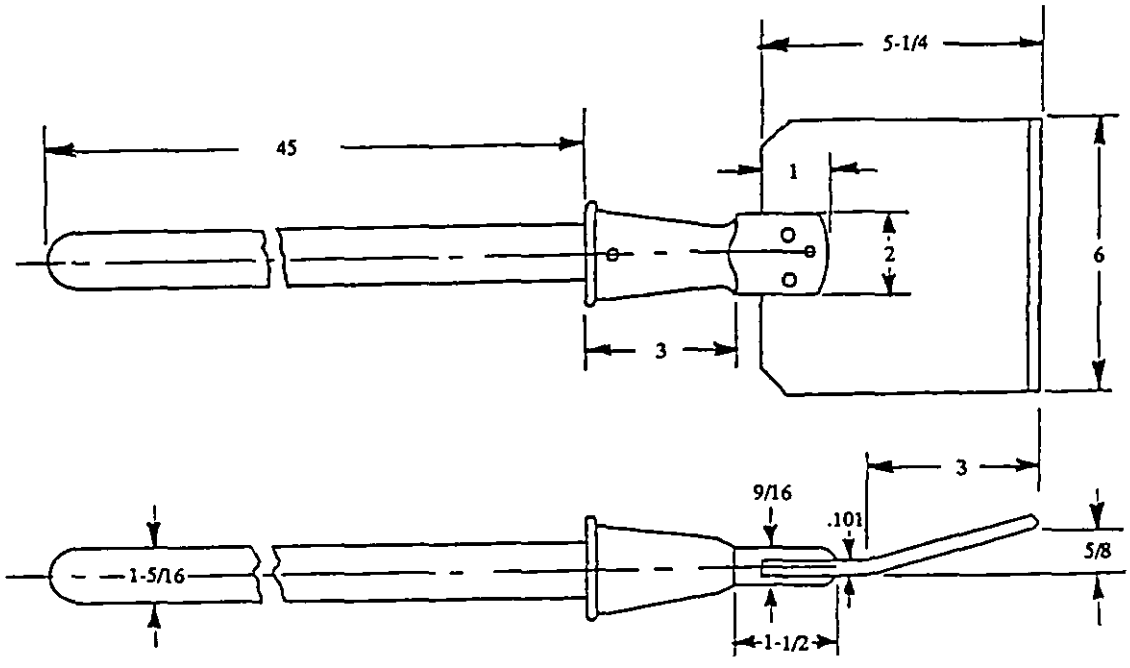


Figure 5. Type IV, ship bottom scraper, wood handle.

3.10.1 Type IV manufacture and design. The scrapers shall have a blade made of steel having a carbon content of 0.40 to 0.50 percent. The blade shall be properly hardened and tempered, and the scraping edge shall be ground to a double-beveled chisel edge. The blade shall rest in a slot and shall be secured by at least three rivets to a steel or malleable iron socket. The socket shall be provided with a taper bore into which the turned taper end of the handle shall be neatly fitted and secured by means of a number 10, 7/8 inch long steel wood screw.

3.10.2 Type IV handle. The handle shall be well proportioned and of a suitable size and shape to provide a comfortable handgrip.

3.10.3 Type IV dimensions. The scrapers shall comply with the dimensions given on figure 5 except that the dimensions may vary within a tolerance of ± 5 percent.

3.11 Type V, ships, triangular scrapers, wood handle. Type V scrapers shall be similar to figure 6.

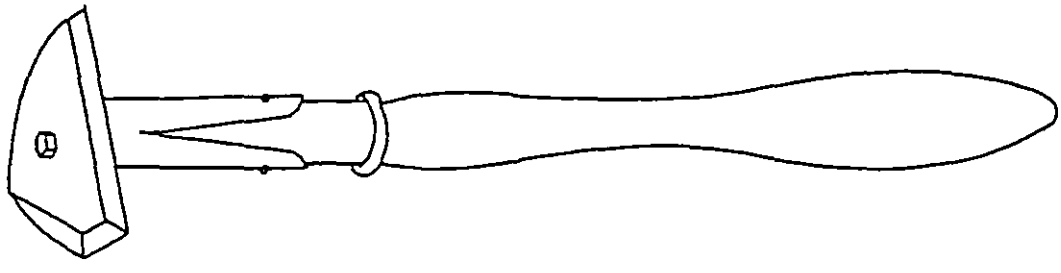


Figure 6. Type V, ships, triangular scraper, wood handle.

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3.11.1 Type V manufacture and design. Type V scrapers shall consist of a blade, a socket, and a wood handle.

3.11.1.1 Type V blade. The blade shall be made of steel having a carbon content of 0.80 to 0.90 percent. It shall be trapezium-shaped, having four scraping edges, each ground at an angle of approximately 60 degrees, and shall be provided with a rectangular hole approximately 3/8 by 7/8 inch in the center. The thickness of the blade shall be minimum 0.122 inch, the lengths of the sides shall be approximately 4-3/4, 4-3/4, 4 inches, and 3/4 inch respectively. The blade shall be secured to the socket by means of a bolt and washer. The bolt shall be 3/4 inch in length, threaded 3/8 inch American National form, and the head shall be either square or hexagonal.

3.11.1.2 Type V socket. The socket shall be of malleable iron, galvanized, and furnished with a straight-grained, well seasoned hickory handle, which shall be inserted in the socket and secured to the handle by means of three or more round-head brass screws. The socket shall conform to the following approximate dimensions: length overall, 5-1/2 inches; outside diameter large end, 1-1/4 inches; small end, 3/4 inch. The small end of the socket shall have a rectangular shoulder about 3/16 inch in height which shall fit the hole in the blade. The shoulder of the socket shall be tapped 3/8 American National tread to suit a 3/8 by 3/4 inch bolt.

3.11.1.3 Type V handle. The wood handle shall be tapered to fit the socket and shall be inserted for a distance of about 4 inches. The large diameter of the handle shall be approximately 1-1/4 inches; the overall length of the complete scraper shall be about 18 inches.

3.12 Workmanship.

3.12.1 Steel. The steel shall be free from burrs, roughness, or excessive oxide scale.

3.12.2 Wood handles. The wood handles shall be smoothly finished.

3.13 Metric product. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within specified tolerances using conversion tables contained in the latest revision of FED-STD-376, and all other requirements of this Federal Specification are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. In accordance with 4.1 above, the supplier is responsible for insuring that components and materials used were manufactured, tested, and inspected in accordance with the requirements of referenced subsidiary specification and standards to the extent specified, or, if none, in accordance with this specification.

4.1.2 Certificates of compliance. When certificates of compliance are submitted, the Government reserves the right to inspect such items to determine the validity of the certification.

4.2 Sampling procedures. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.2.1 Visual examination. Ship scrapers shall be examined for the defects listed in table I. The lot size shall be expressed in terms of ship scrapers. The sample unit shall be one scraper. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 4.0.

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TABLE I. End item visual defects

Examine	Defect
Construction and Workmanship	Not type, class specified Wood handles not well proportioned; not a comfortable grip; not suitable shape Wood handles not ferruled, ends not tapered Blades not shape, size specified; not secured to socket; edges not ground; securing bolt threads nonconforming
Material	Not material specified Wood handles not straight-grain hickory, maple or ash
Finish	Steel not painted, finished, or preserved as required Wood handles not covered with protective coating where specified Steel not free of burrs, rough spots, oxide scale; handles not smooth finish
Marking	Not marked as specified; marking not permanent; not legible; data incomplete

4.2.2 Dimensional examination. The ship scrapers shall be examined for any nonconformance to dimensional requirements. The lot size shall be expressed in terms of ship scrapers. The sample unit shall be one ship scraper. The inspection level shall be S-4 and the AQL, expressed in terms of defects per hundred units, shall be 1.5.

4.2.3 Carbon content examination (certification). The supplier shall furnish certification that the chemical composition of each lot or batch of steel used in the end item meets the applicable carbon requirements as specified in 3.6.1, 3.8.1, 3.10.1, and 3.11.1.1.

4.3 Test methods. The ship scrapers shall be tested for the characteristics specified in 4.3.1. The lot size shall be expressed in units of ship scrapers. The sample unit shall be one ship scraper. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 1.0.

4.3.1 Hardness test. Samples of each type and class of scrapers shall be tested in accordance with ASTM E 18 for determination of compliance with hardness requirements specified in applicable paragraphs in section 3.

4.4 Inspection of preparation for delivery requirements. An inspection shall be made to determine that the preservation, packaging, packing, and marking comply with the requirements in section 5 of this specification. Defects shall be scored in accordance with table II. The lot size shall be the number of shipping containers in the end item inspection lot. The sample unit shall be one container. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

TABLE II. Classification of preparation for delivery defects

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; improper size, location, sequence, or method of application
Materials	Any component missing, damaged, or not as specified

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TABLE II. Classification of preparation for delivery defects (cont'd)

Examine	Defect
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling Distortion of container
Contents	Number per container is more or less than required Net weight exceeds requirements

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking shall be in accordance with PPP-P-40. The level of preservation, packaging, and packing shall be A, B, or Commercial, as specified (see 6.1).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Type and class specified (see 1.2).
c. Specify carbon certification (see 3.2.1), if required.
d. Level of packaging and packing required (see 5.1).

MILITARY INTERESTS:

Military Coordinating Activity

Army - GL

Custodians

Army - GL
Air Force - 99

Review Activities

Army - GL
Air Force - 84

User Activity

Navy - MC

CIVIL AGENCY COORDINATING ACTIVITY:

GSA - FSS

PREPARING ACTIVITY:

GSA - FSS

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